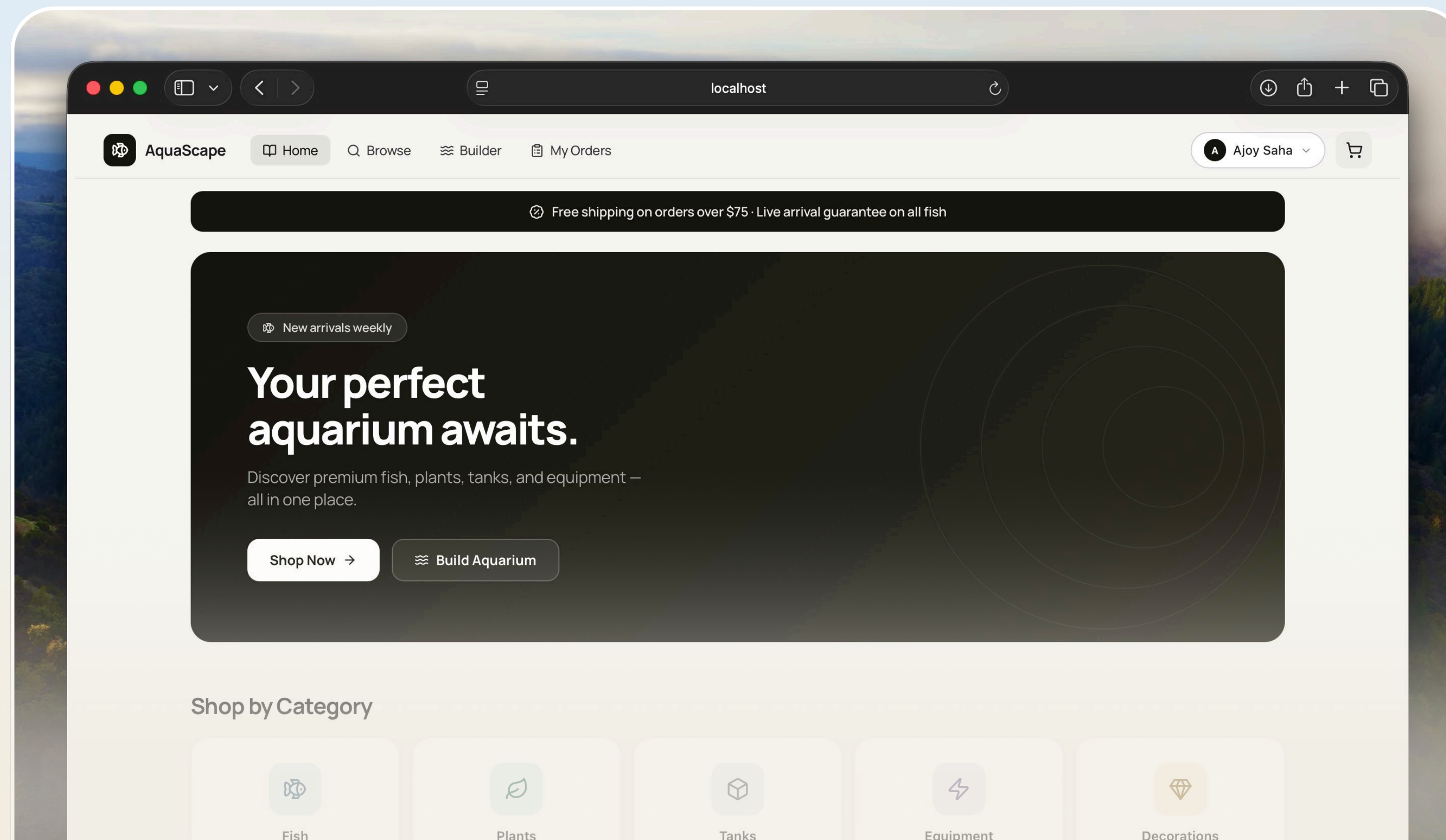


AquaScape

AquaScape is a full-stack web application built for an aquarium retail business



Key Features

Aquarium Compatibility

Checks water type, temperature, tank capacity and species conflicts through database triggers and functions.

Order Processing in PL/SQL

One procedure turns a cart into an order, deducts stock, applies discounts and clears the cart as a single atomic transaction.

Automatic Stock Control

Triggers deduct inventory on every sale, block negative stock, and raise low stock alerts.

Secure Accounts

Customers register and sign in with email and password; passwords are stored as salted script hashes.

Aquarium Builder

Customers design a full tank step by step — choose a tank, add livestock and equipment, validate it.

Live Reporting Views

Thirteen database views feed the dashboard and reports — KPIs, monthly sales, best sellers, profit and stock movement.

Technology Stack

React → Express → Oracle → Docker volume

Database Architecture

Schema Overview

- 32 tables across 4 functional domains
- Domain tables:
Core (12) • Builder (7) • Order (7) • Supplier (6)
- ER diagram represents the complete data model

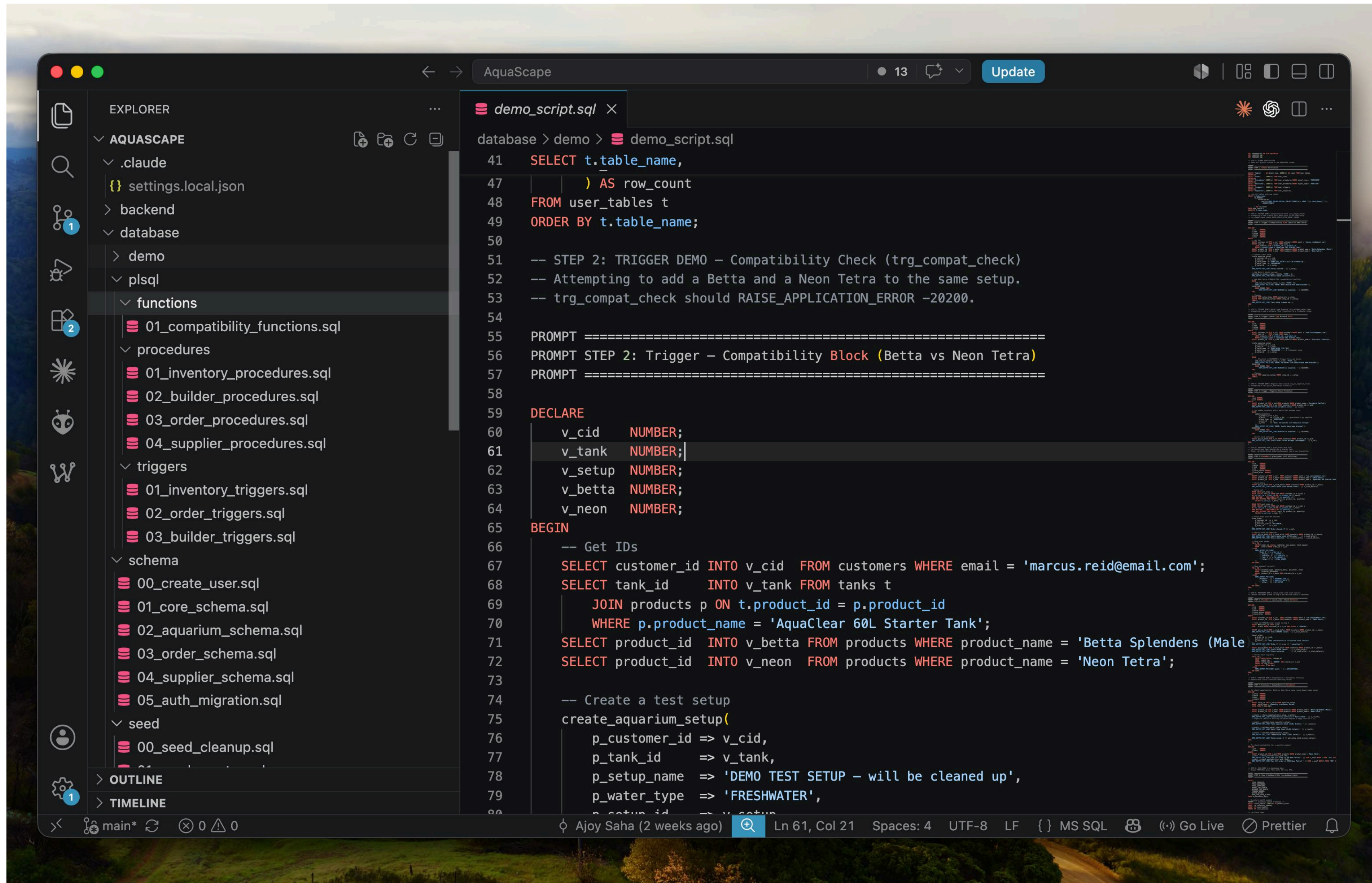
Stored Procedures

- 14 procedures — all business logic runs inside Oracle
- Example: *place_order*
Insert Order → Copy Cart Lines →
Stock Deducted → Cart Cleared → Commit
- Business rules and validations implemented in procedures

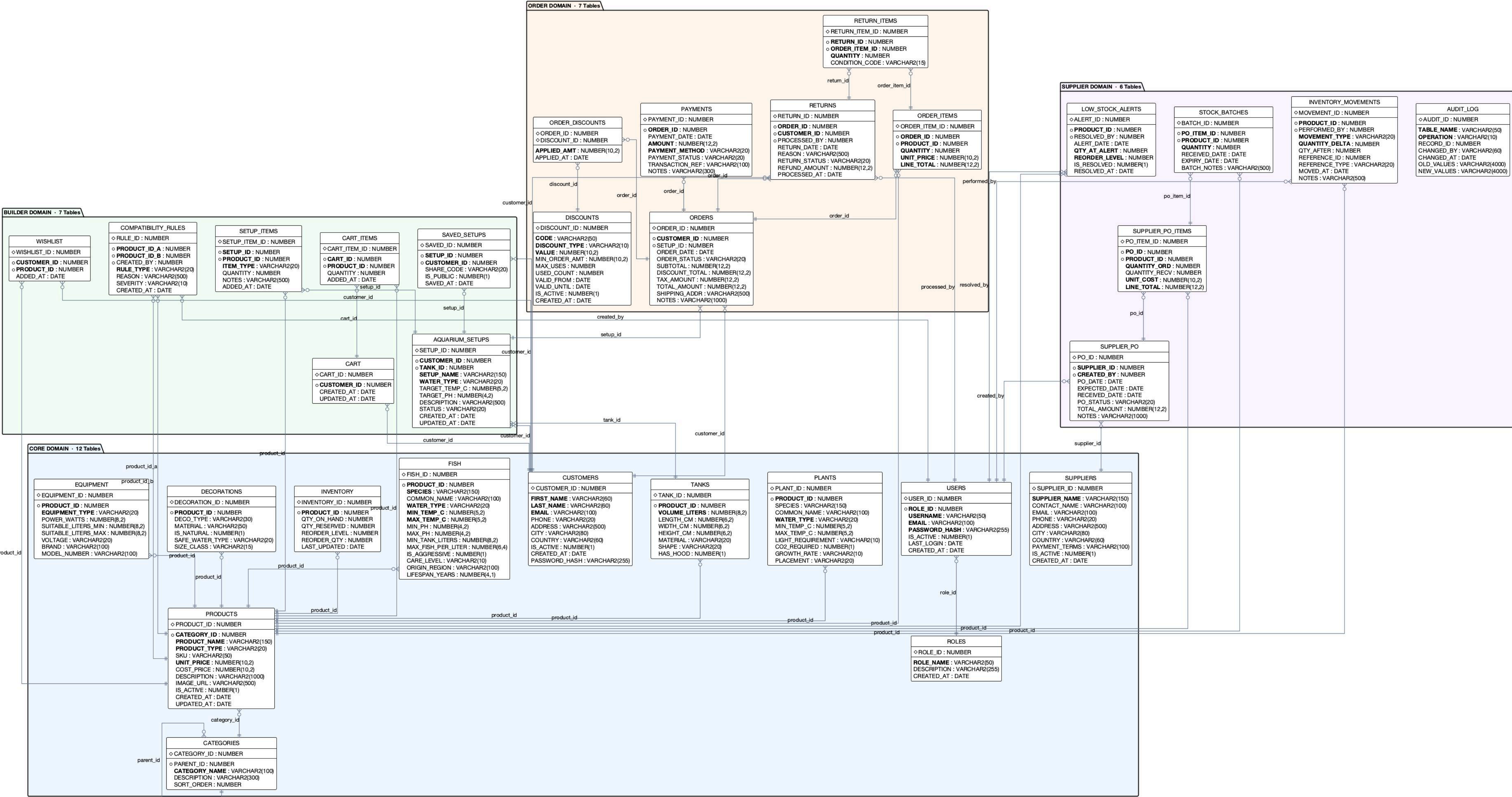
Triggers

- 16 triggers — rules that fire no matter who writes
- Grouped into 4 buckets:
Stock Protection • Alerts • Compatibility • Audit
- Demo hook: SQL*Plus rejects raw brackish-puffer INSERT with ORA-20201

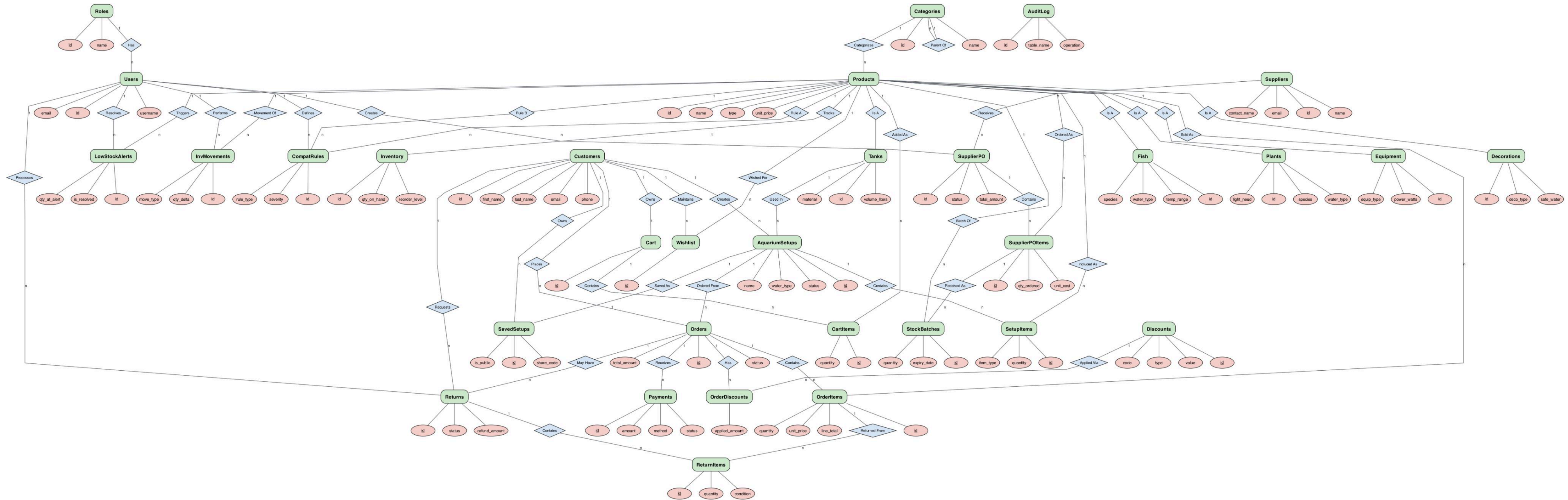
Database PL/SQL, Triggers, Schema



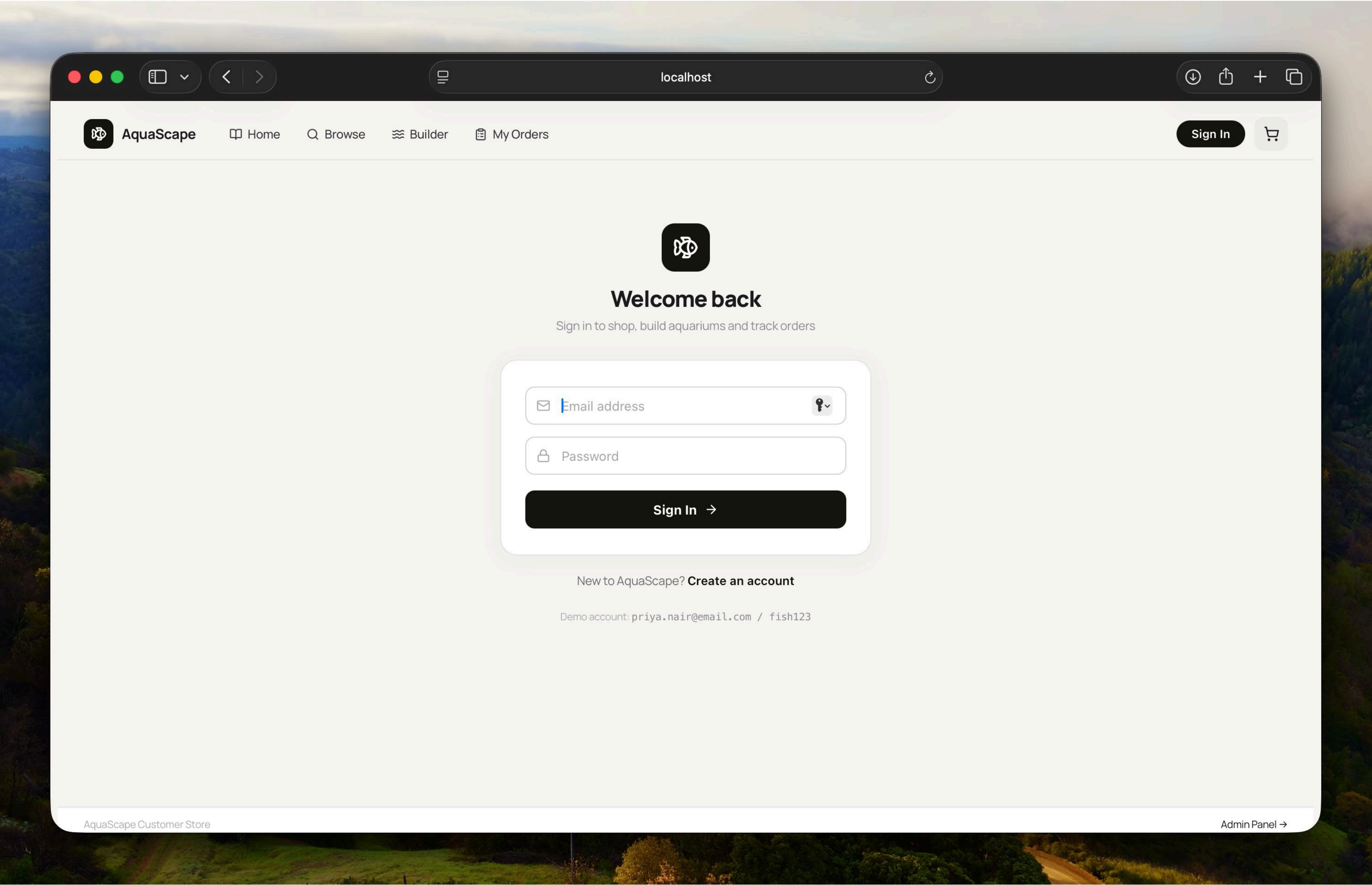
Schema Diagram

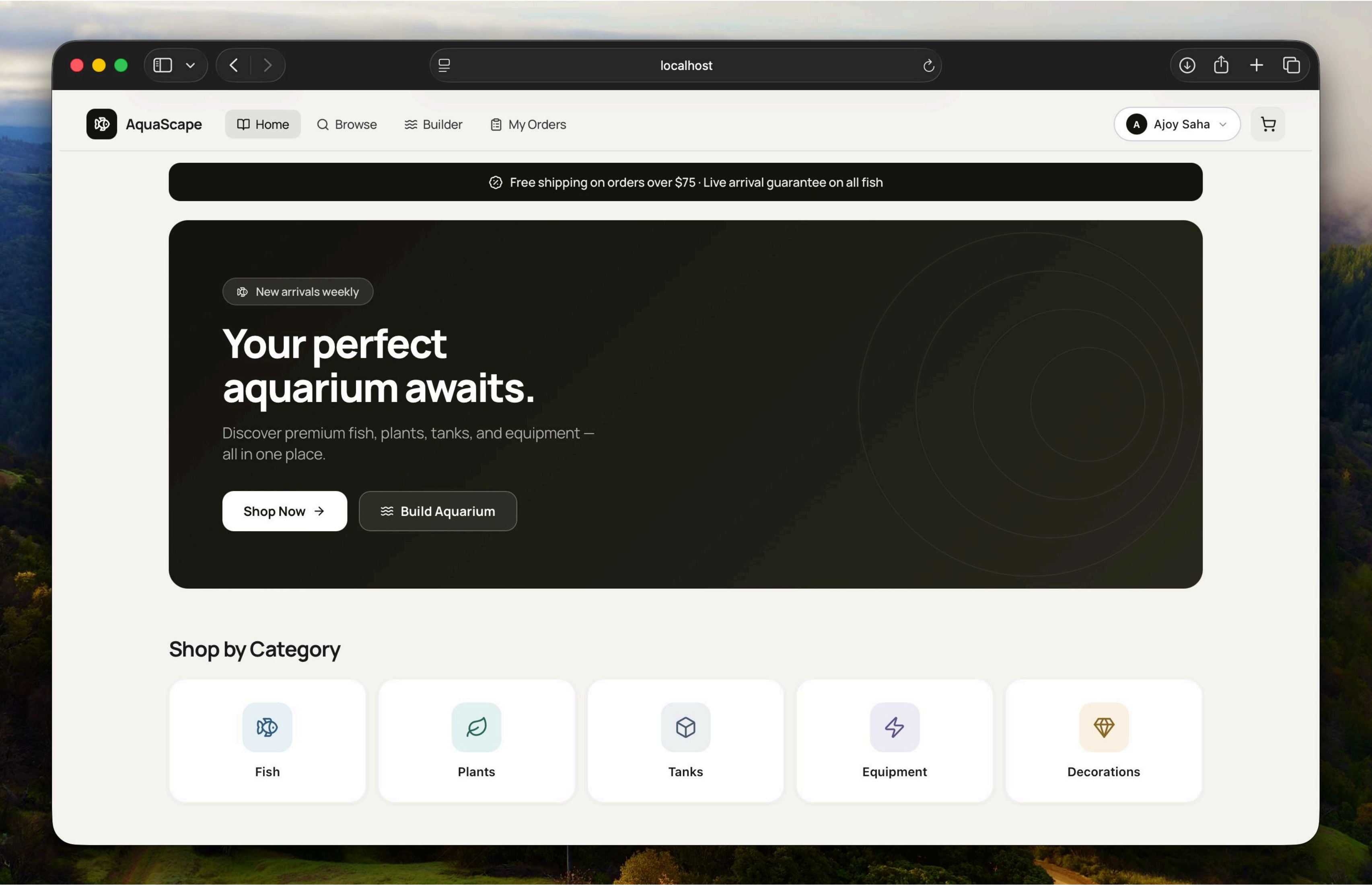


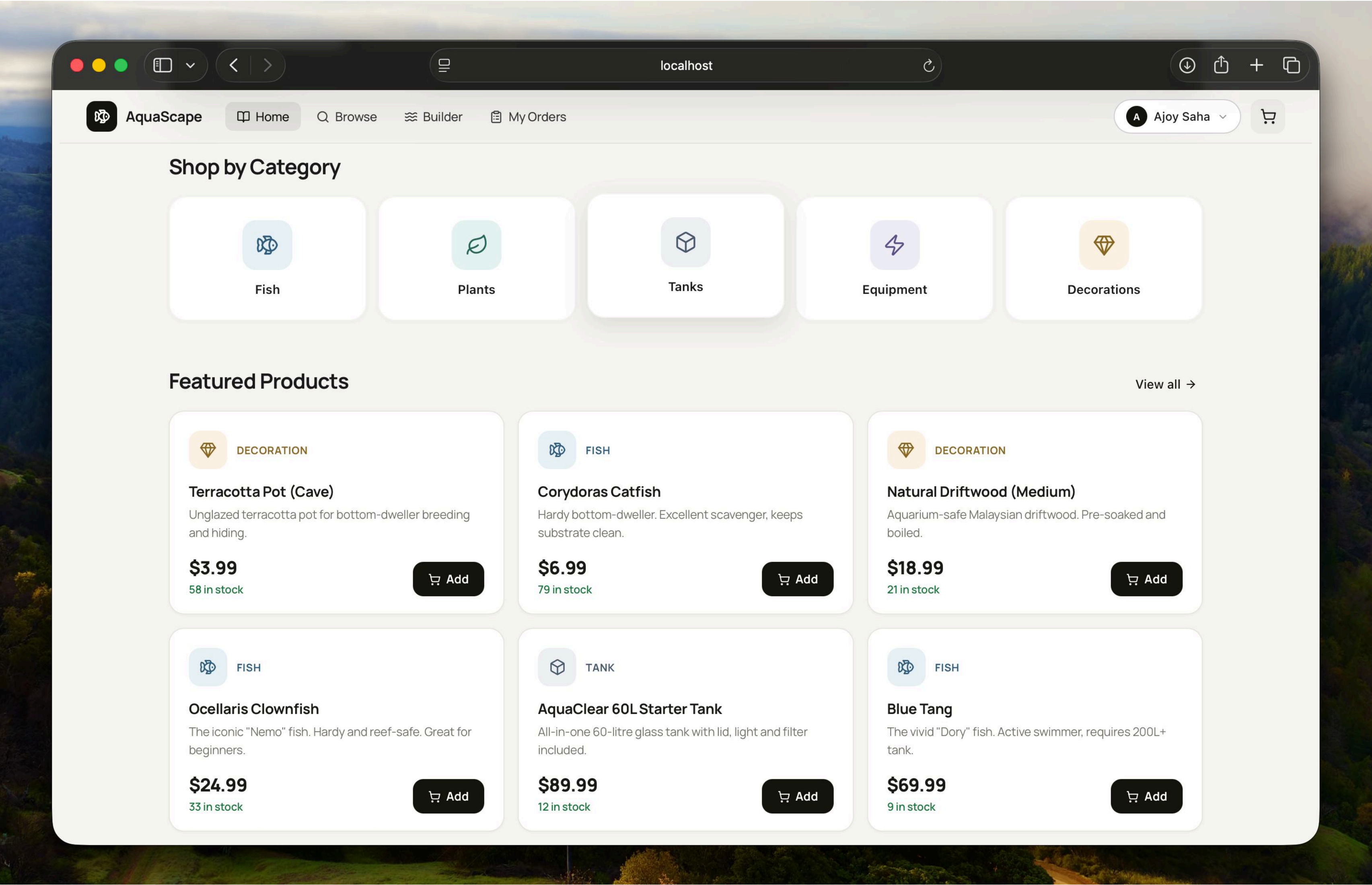
ER Diagram

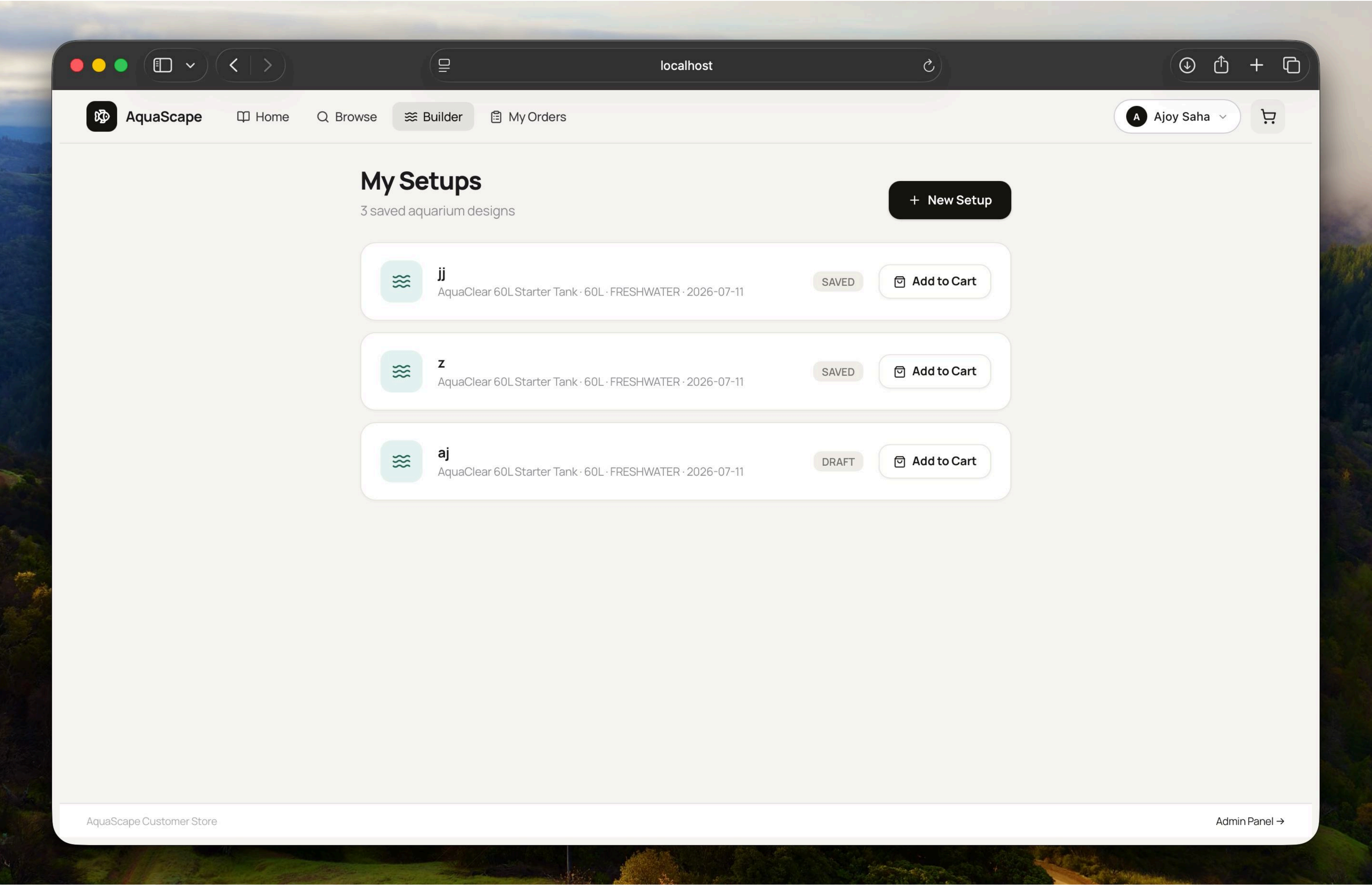


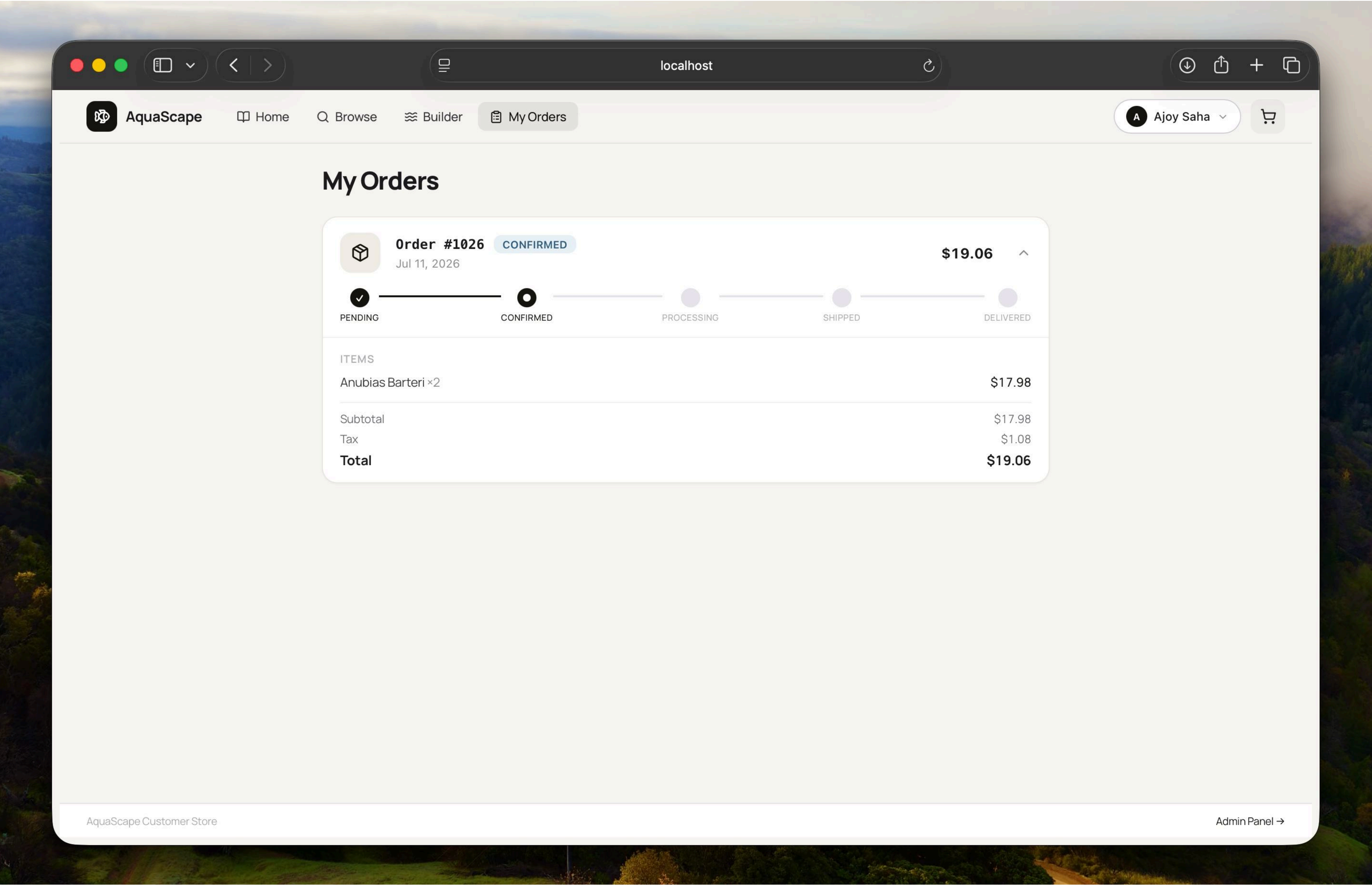
Project Screenshots

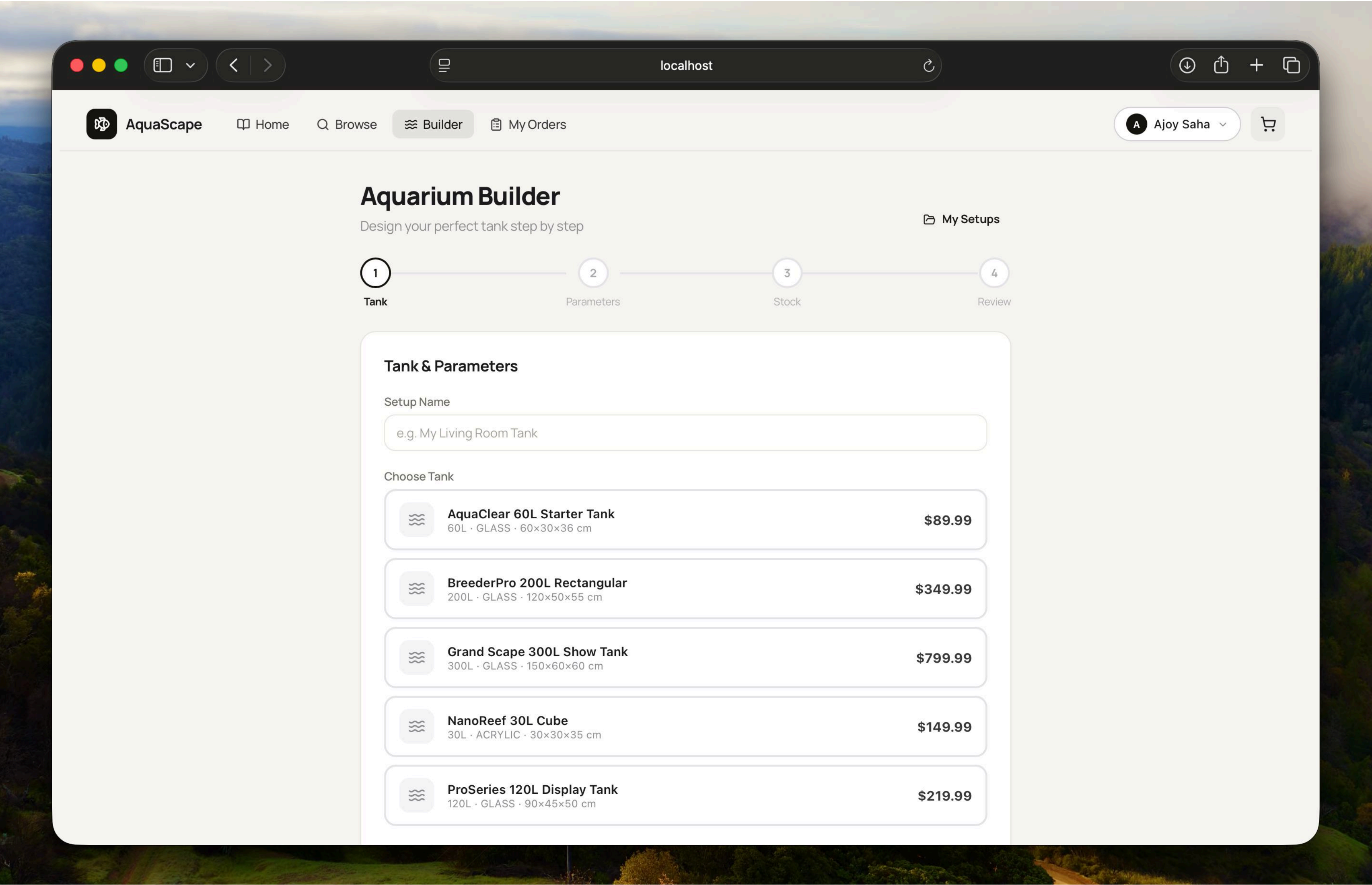


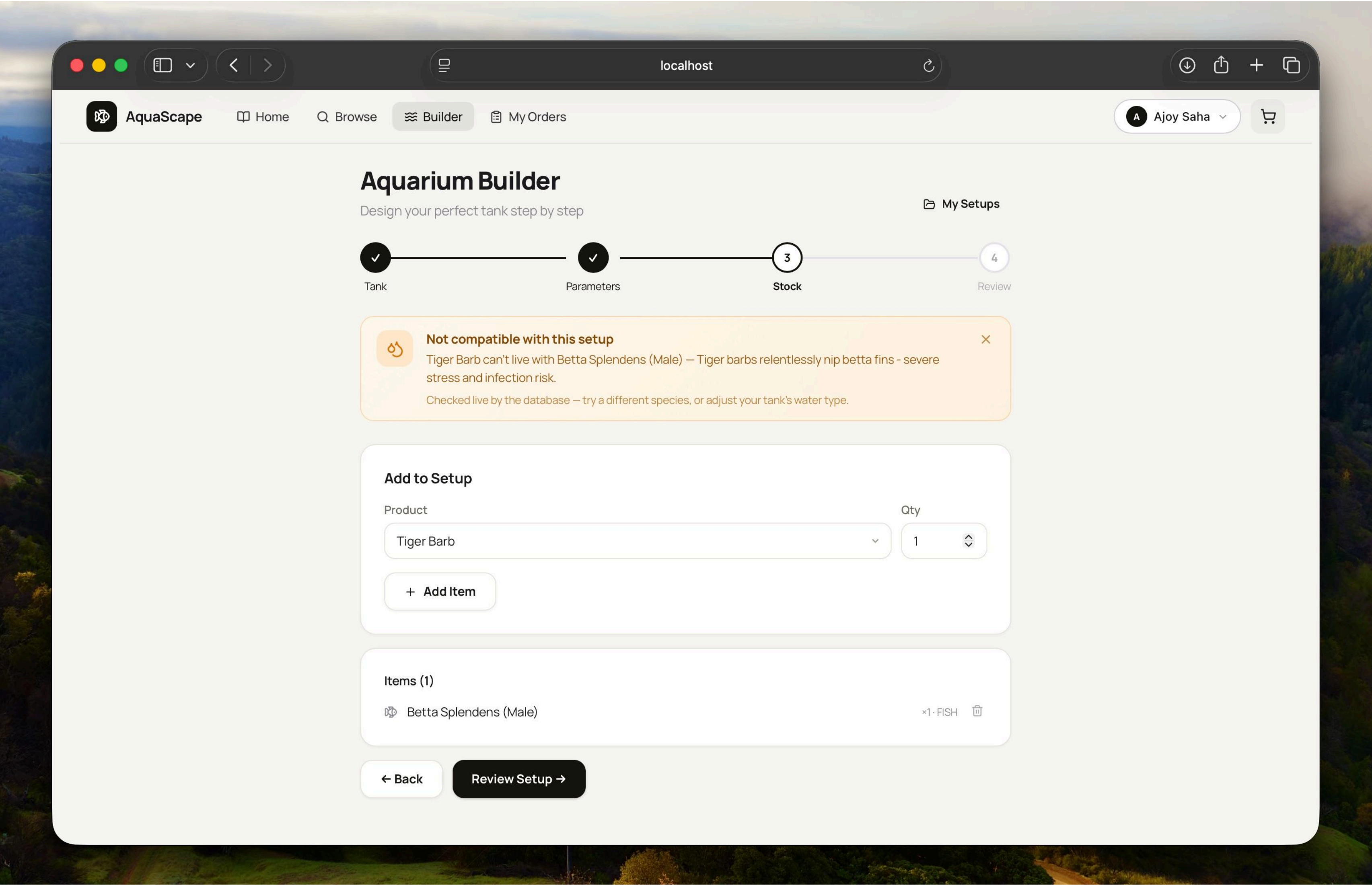


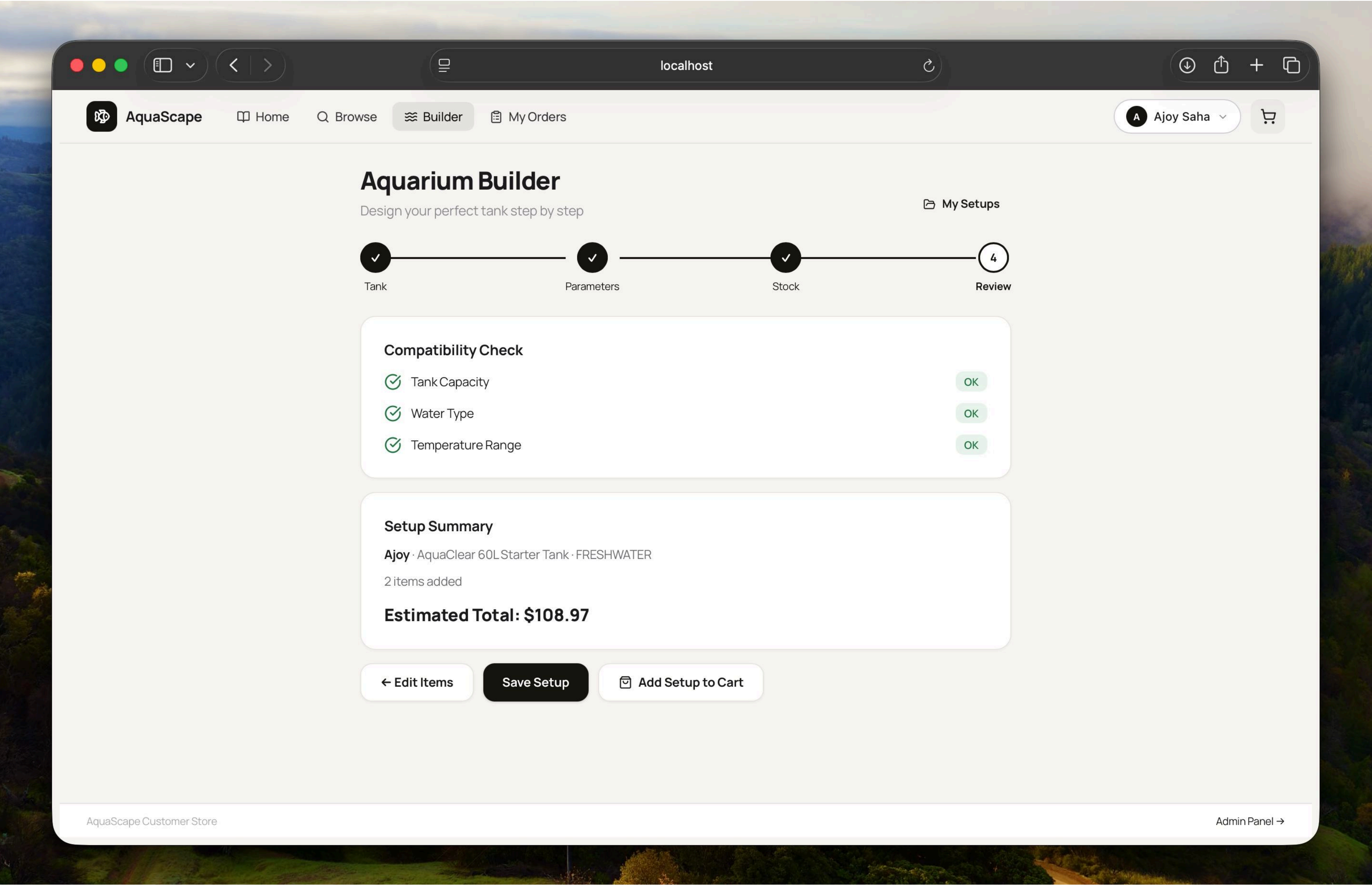


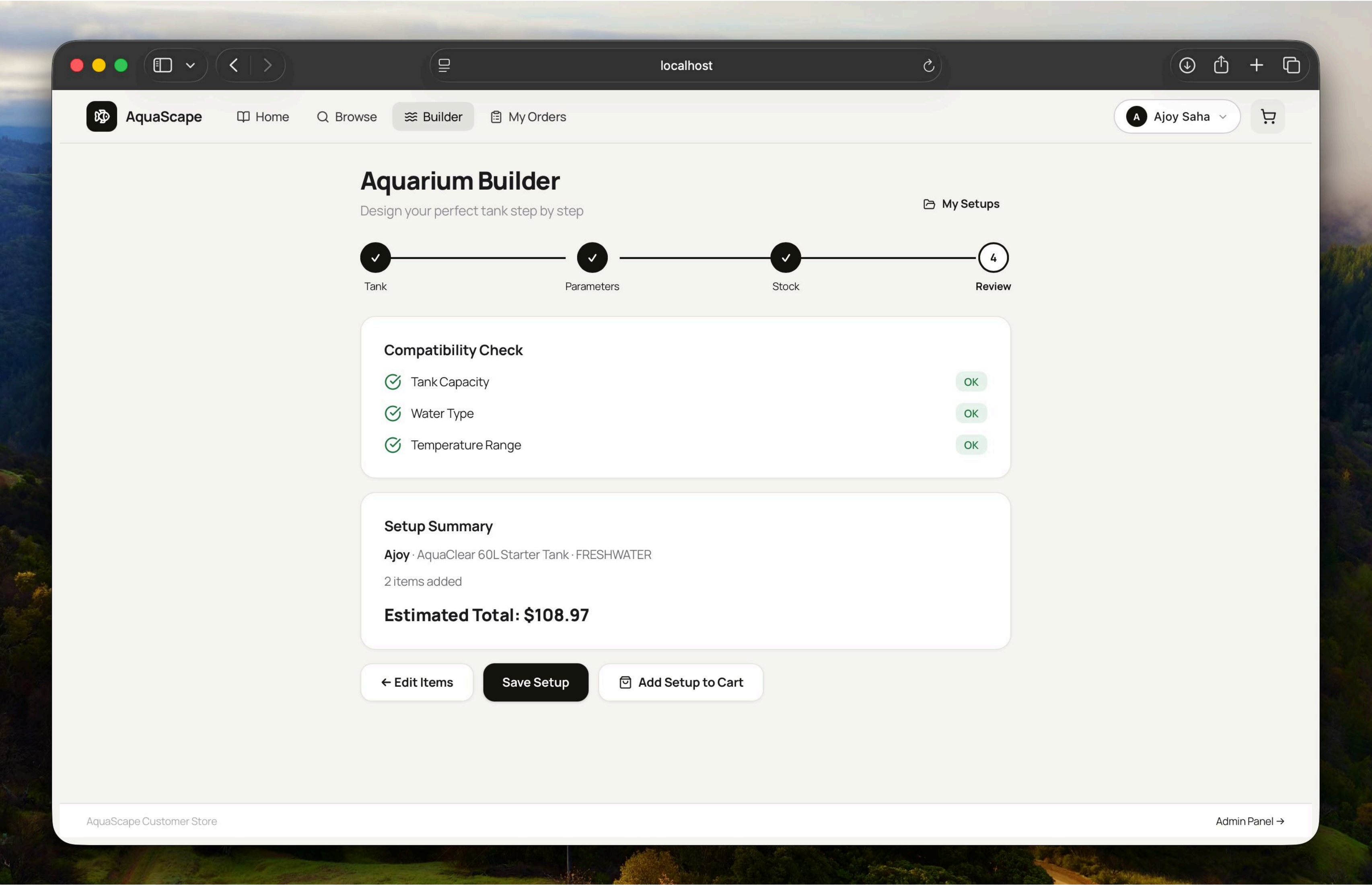


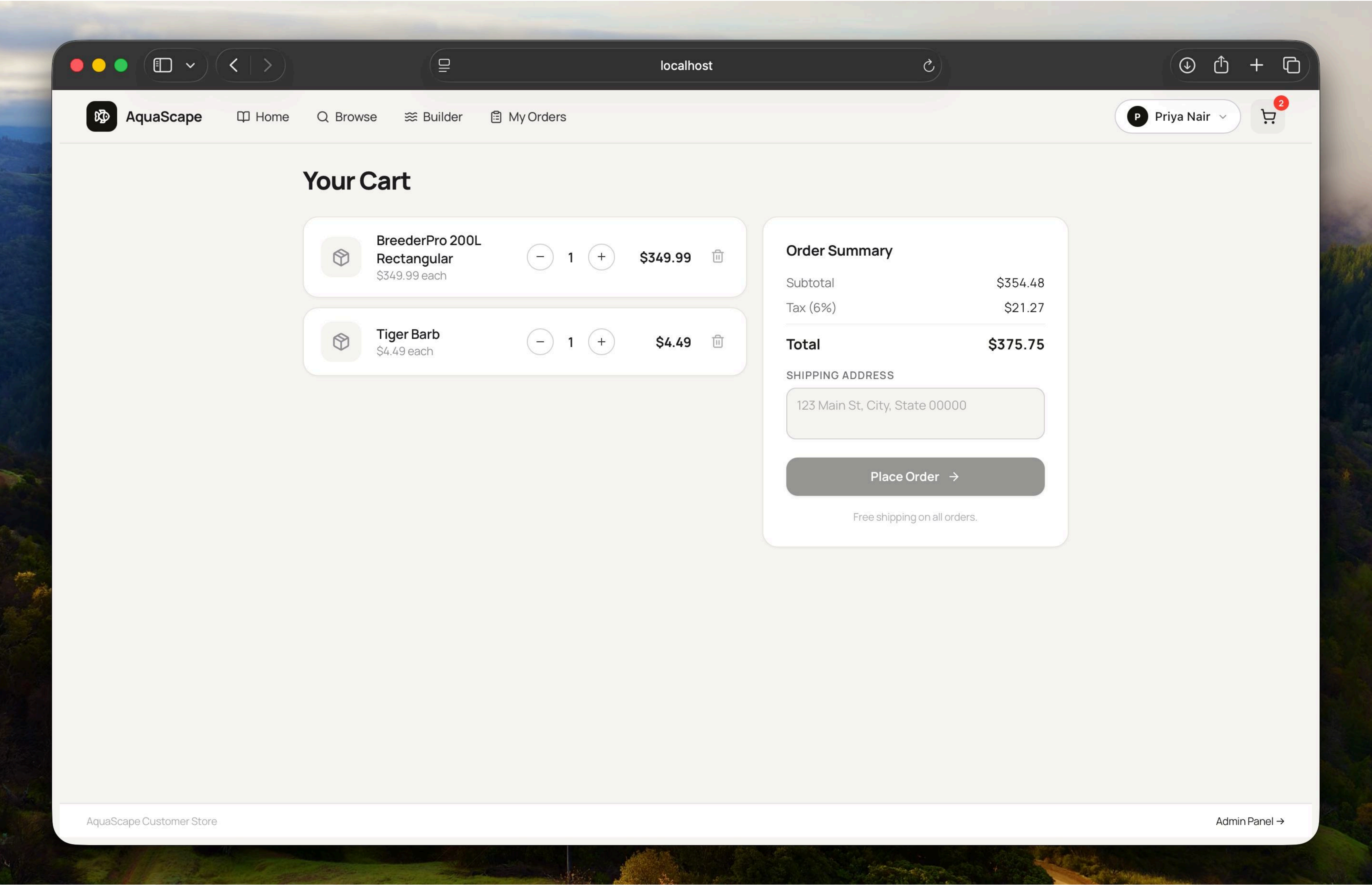


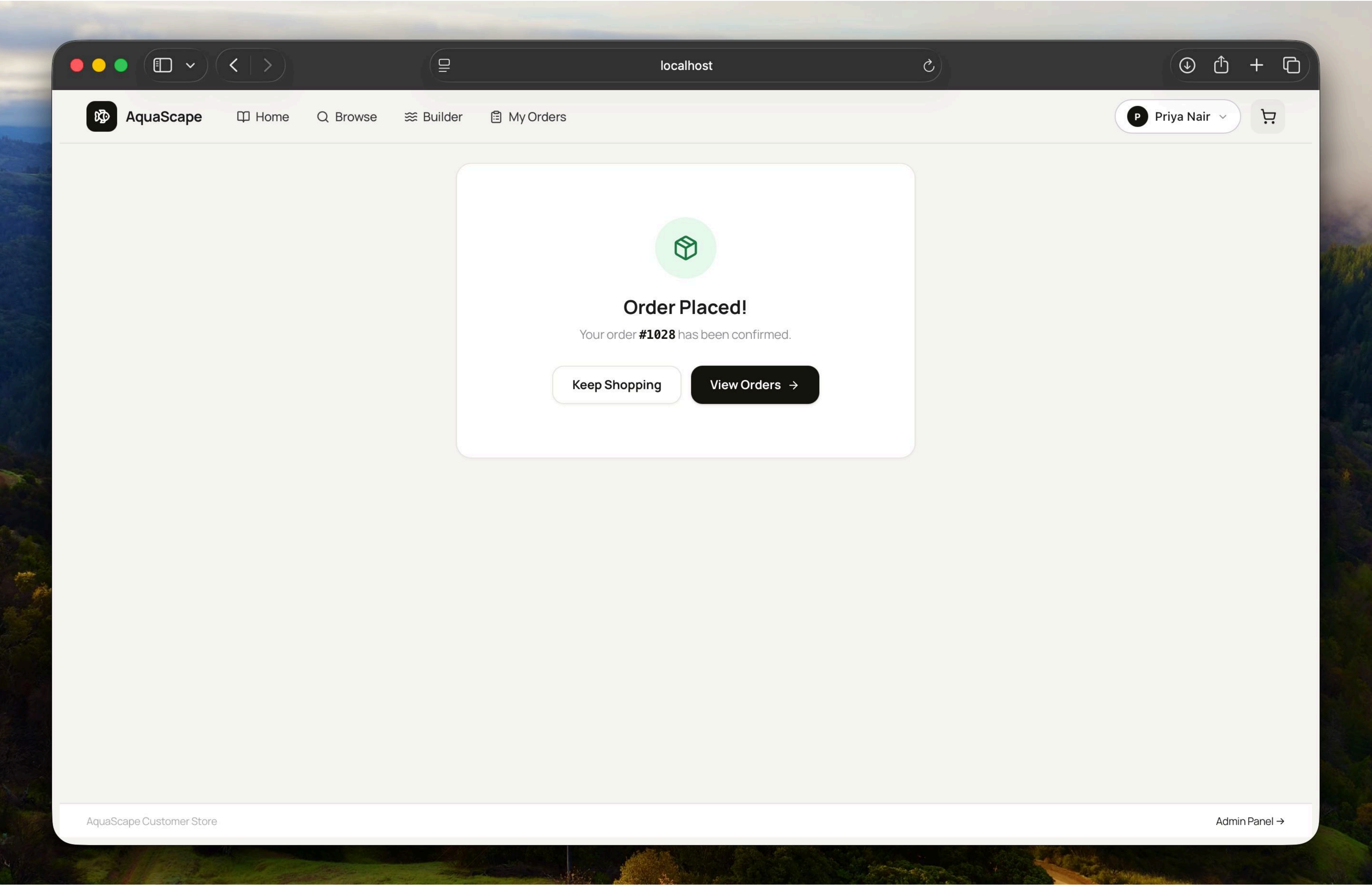






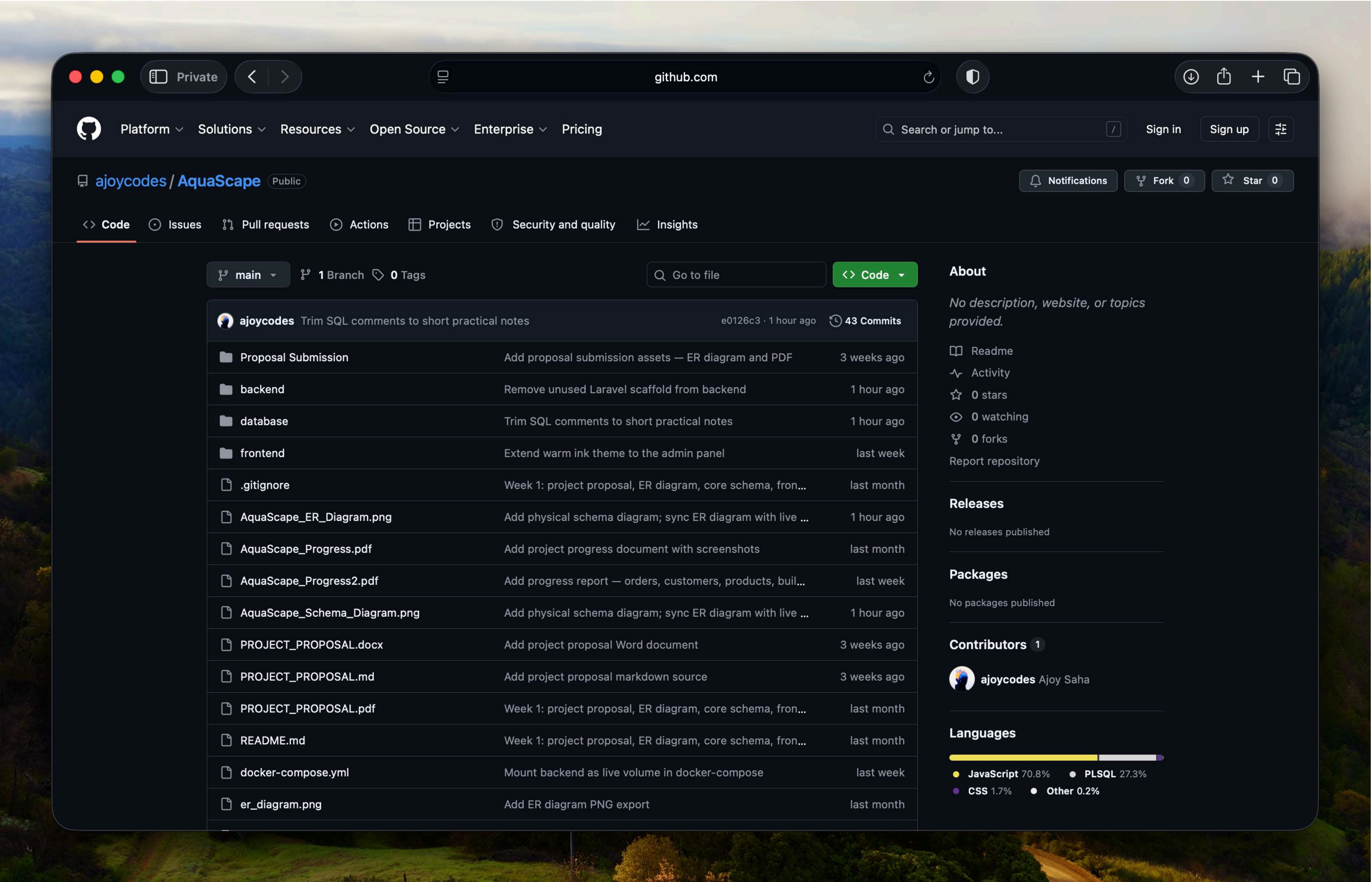






Github

Repository



Thank you!